

Aligning Reconstruction with Generation: A Latent Distribution Alignment Perspective on Evaluation and Optimization

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Abstract

In latent generative models, reconstruction quality is often assumed to correlate with generative performance. However, reconstruction FID (rFID) exhibits weak or even negative correlation with generation FID (gFID). We show that this discrepancy arises from a systematic misalignment: Reconstruction evaluates the decoder on encoder latents, whereas generation operates on generative latents drawn from a different distribution. This latent distribution mismatch leads to a pronounced gap between reconstruction quality and generative performance. To address this issue, we introduce generation-aware reconstruction (GAR), which aligns reconstruction with the generative process by perturbing encoder latents with noise and denoising them using the generative model before decoding. From a latent distribution perspective, we show that GAR evaluates the decoder along the generative trajectory, bridging the encoder and generative latent distributions. Empirically, GAR-FID exhibits very strong empirical correlation with gFID across a wide range of architectures and model scales, in contrast to standard reconstruction metrics. Building on this insight, we further propose decoder adaptation, which trains the decoder on intermediate generative latents, leading to enhanced generative quality. Our results suggest that latent distribution alignment provides a useful unifying perspective for both evaluating and improving latent generative models.

1 Introduction

Latent generative models have become a dominant paradigm for high-quality image synthesis, typically combining a visual tokenizer with a latent generative model such as diffusion or flow matching [Rom+22; Pre+22; PX22; AVE23; LGL23; Gen+25a]. Within this framework, reconstruction quality is often implicitly assumed to reflect generative performance: a better decoder is expected to yield both better reconstructions and better generated samples [OVK17; ERO21; Rom+22; Pre+22]. However, recent empirical studies [YW25; Kou+25; Ye+25; Sko+25; Che+25b; Xu+26] challenge this assumption. In practice, reconstruction FID (rFID) frequently exhibits weak or even negative correlations with generation FID (gFID), a phenomenon often referred to as the *reconstruction–generation dilemma*. This raises a fundamental question: *why does reconstruction fail to predict generative performance?*

In this work, we argue that this discrepancy is not incidental, but can be attributed to a structural misalignment between reconstruction and generation. Specifically, reconstruction evaluates the decoder on encoder latents ($z_e \sim \mathcal{P}_e$), while generation operates on generative latents obtained via diffusion or flow matching models [Rom+22; Pre+22; LGL23; Gen+25a], which follow a different

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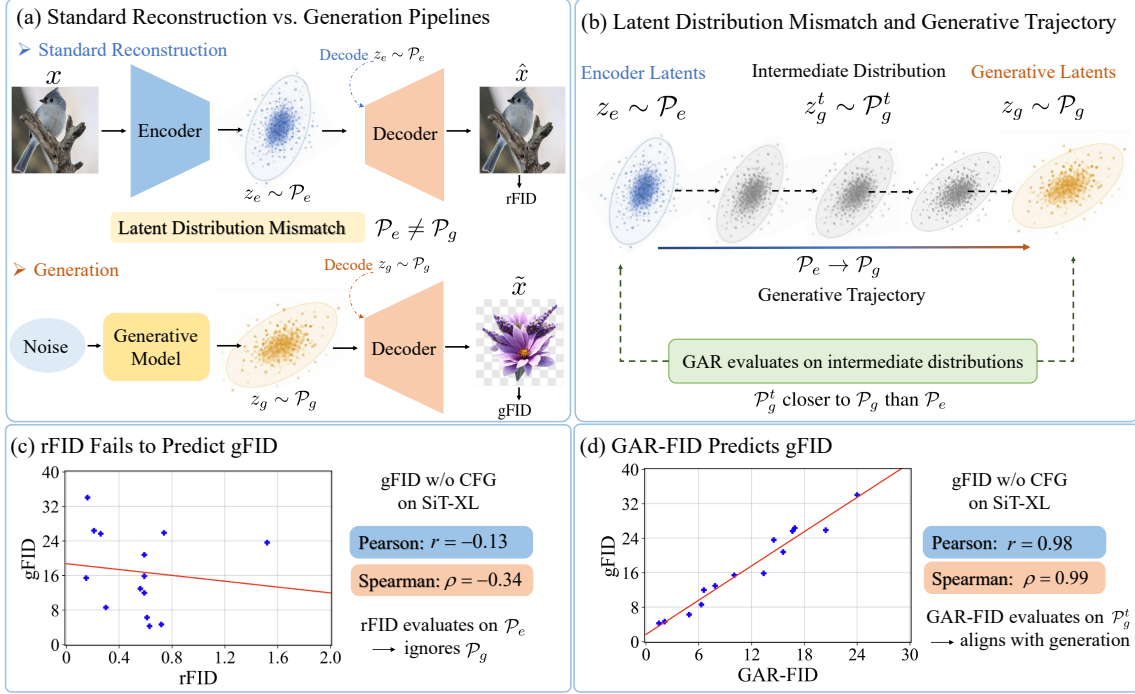


Figure 1: Generation-aware reconstruction resolves the reconstruction-generation mismatch.

(a) Standard reconstruction and generation operate on different latent distributions: reconstruction decodes encoder latents $z_e \sim \mathcal{P}_e$, while generation decodes generative latents $z_g \sim \mathcal{P}_g$, leading to a mismatch. (b) The generative process induces a trajectory of intermediate latent distributions $\mathcal{P}_g^t$ that progressively bridge $\mathcal{P}_e$ and $\mathcal{P}_g$. (c) As a result, standard reconstruction metrics (rFID) are poorly correlated with generative performance (gFID), with Spearman rank correlation $\rho = -0.34$. (d) In contrast, generation-aware reconstruction (GAR-FID), which evaluates on intermediate latents z_g^t , exhibits strong correlation with gFID at $\eta_t = 0.8$, achieving $\rho = 0.99$.

latent distribution ($z_g \sim \mathcal{P}_g$), as illustrated in Figure 1(a). As a result, rFID measures decoder behavior under a distribution that is not encountered at generation time, leading to a systematic gap between reconstruction quality and generative performance.

We show that this gap can be naturally understood through a latent distribution perspective. By analyzing the distributions of encoder latents and generative latents, we reveal a structured mismatch that is strongly associated with generative performance. In particular, as shown in Figure 1(b), we identify a continuous trajectory of latent distributions induced by the generative process, which progressively bridges the encoder distribution and the generative distribution. This perspective provides a coherent explanation for the failure of standard reconstruction metrics and highlights latent distribution mismatch as an important factor underlying the reconstruction-generation gap.

Building on this insight, we introduce *generation-aware reconstruction* (GAR), a simple yet effective modification of the reconstruction process. Instead of directly decoding encoder latents, GAR perturbs them with noise and passes them through the generative model before decoding. This procedure evaluates the decoder under generation-consistent intermediate latent distributions, effectively aligning reconstruction with the generative process. As a result, the corresponding metric, GAR-FID, exhibits very strong empirical correlation with gFID across a wide range of architectures and model scales, in stark contrast to standard reconstruction metrics, as shown in Figure 1(c) and (d).

Beyond evaluation, our analysis suggests a broader perspective: *latent distribution alignment is important for both evaluating and improving latent generative models*. If generative performance depends on how well the decoder operates on generative latents, then improving generation can benefit from explicitly aligning the decoder with these distributions. Motivated by this insight, we

further propose *decoder adaptation*, which fine-tunes the decoder on intermediate generative latents. This simple strategy effectively adapts the decoder for the distributions encountered during generation, leading to improved generative performance. We also observe that this alignment mitigates visual artifacts and hallucination-like inconsistencies, yielding more semantically consistent samples.

Overall, our work suggests that latent distribution alignment provides a unifying perspective that connects evaluation and optimization in latent generative models. By revealing the role of distribution mismatch and providing an effective way to address it, we bridge the gap between reconstruction and generation, and offer a new perspective for designing and improving generative systems.

Contributions. Our contributions are summarized as follows:

1. We identify an **evaluation misalignment** in latent generative models: reconstruction evaluates the decoder on encoder latents, whereas generation operates on a different latent distribution.
2. We provide a **latent distribution perspective** that explains this misalignment, revealing a generative trajectory from $\mathcal{P}_e$ to $\mathcal{P}_g$ and demonstrating a strong empirical association between latent mismatch and generative quality.
3. We introduce **generation-aware reconstruction (GAR)**, a simple yet well-motivated evaluation protocol that aligns reconstruction with the generative process, and show that GAR-FID exhibits strong empirical correlation with gFID, while rFID does not.
4. Motivated by this insight, we propose **decoder adaptation**, which aligns decoder training with intermediate generative latents and improves GAR and downstream generative quality.

2 Revisiting Reconstruction and Generation

We revisit the standard reconstruction and generation pipelines and identify a systematic mismatch between them due to differing latent distributions. To address this discrepancy, we introduce generation-aware reconstruction (GAR), which makes the input distributions more consistent.

2.1 Standard Reconstruction and Its Misalignment with Generation

Latent visual generative models [Rom+22] are typically decomposed into two independently trained components: (i) a continuous visual tokenizer [Che+25c; Che+25b; Yan+26; Kou+25], and (ii) a latent diffusion or flow-matching model [Pre+22; PX22; AVE23; LGL23; Gen+25a]. This decomposition induces two pathways for image synthesis: standard reconstruction and generation.

As depicted in Figure 1(a), the standard reconstruction pipeline maps an input image x to a latent representation $z_e = \mathcal{E}_\theta(x)$, inducing a latent distribution $z_e \sim \mathcal{P}_e$. A decoder $\mathcal{D}_\phi$ reconstructs the image as $\hat{x} = \mathcal{D}_\phi(z_e)$, and reconstruction quality is evaluated using the Fréchet Inception Distance [FID; Heu+17], referred to as rFID. In contrast, the generation pipeline operates on generative latents. Starting from noise $\varepsilon \sim \mathcal{N}(0, I)$, a generative model $\mathcal{G}_\varphi$ produces latents $z_g = \mathcal{G}_\varphi(\varepsilon) \sim \mathcal{P}_g$, which are decoded as $\tilde{x} = \mathcal{D}_\phi(z_g)$. Generative quality is measured by FID between real and generated images, referred to as gFID.

This distinction reveals the core misalignment: reconstruction evaluates the decoder on inputs drawn from $\mathcal{P}_e$, whereas generation evaluates it on inputs drawn from $\mathcal{P}_g$. As a result, rFID reflects decoder behavior under $\mathcal{P}_e$, while gFID depends on its behavior under $\mathcal{P}_g$. Therefore, there is no reason to expect one to predict the other. This suggests that the failure of standard reconstruction metrics, such as rFID, arises from an evaluation misalignment between $\mathcal{P}_e$ and $\mathcal{P}_g$.

2.2 Generation-Aware Reconstruction

To better align reconstruction evaluation with generative performance, a meaningful reconstruction metric should evaluate the decoder under the latent distributions encountered during generation. To this end, we introduce *generation-aware reconstruction* (GAR), which aligns reconstruction with the

generative pipeline. As illustrated in Figure 2, instead of directly decoding encoder latents z_e , GAR first perturbs them with noise following the forward process of the generative model. Specifically, given $z_e \sim \mathcal{P}_e$, we construct noisy encoder latents $z_{\text{noisy}}^t = a_t z_e + b_t \epsilon$, where a_t and b_t are noise scheduling coefficients at timestep t . We define the noise level as $\eta_t = \frac{b_t}{a_t + b_t}$, which controls the relative contribution of noise and encoder latents. This formulation reveals that GAR continuously interpolates between reconstruction and generation. When $\eta_t = 0$, we recover standard reconstruction with $z_{\text{noisy}}^t = z_e$; when $\eta_t = 1$, the process corresponds to generation from pure noise ϵ . Intermediate values of η_t bridge these two regimes. The generative model $\mathcal{G}_\varphi$ is then applied to denoise z_{noisy}^t , producing latents $z_g^t = \mathcal{G}_\varphi(z_{\text{noisy}}^t) \sim \mathcal{P}_g^t$, which are decoded as $\bar{x} = \mathcal{D}_\phi(z_g^t)$.

By passing encoder latents through the noise–denoise process, GAR progressively shifts them toward the generative distribution $\mathcal{P}_g$. In practice, the resulting latents follow intermediate distributions $\mathcal{P}_g^t$, which are closer to $\mathcal{P}_g$ than to $\mathcal{P}_e$. These distributions can be interpreted as points along the generative trajectory, providing a continuous path from encoder latents to the generative distribution. From this perspective, GAR can be viewed as an evaluation protocol that aligns reconstruction with generation by evaluating the decoder under intermediate generative latent distributions. Accordingly, GAR reframes the role of reconstruction from measuring fidelity under encoder latents to evaluating decoder behavior under generation-consistent intermediate latents. Based on this formulation, we define a generation-aware reconstruction metric (GAR-FID), computed as the Fréchet Inception Distance between original images and their GAR reconstructions $\bar{x}$. As we will demonstrate empirically, standard reconstruction metrics such as rFID fail to reliably predict generative performance, whereas GAR-FID significantly improves this predictive relationship.

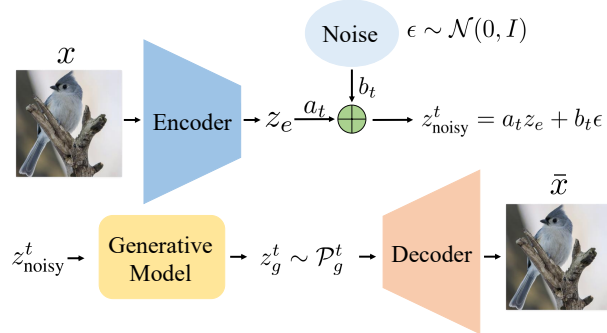


Figure 2: Illustration of the GAR pipeline.

3 Latent Distribution Mismatch and the Generative Trajectory

In this section, we conduct a systematic empirical analysis of latent distribution mismatch and its implications for generation-aware reconstruction (GAR). We begin by quantifying the discrepancy between the encoder-induced distribution $\mathcal{P}_e$ and the generative distribution $\mathcal{P}_g$. We then show that GAR progressively shifts latents toward $\mathcal{P}_g$ along the generative trajectory. Finally, we examine how latent distribution mismatch relates to downstream generative performance. Additional analysis of structural and perceptual reconstruction quality under GAR is provided in Appendix C.

3.1 Latent Distribution Mismatch

To characterize the distribution mismatch between encoder latents z_e and generative latents z_g , we conduct a systematic empirical analysis using the SD-VAE [Rom+22] encoder and iMF [Gen+25b] as representative models. Since each latent is high-dimensional ($z_e \in \mathbb{R}^{32 \times 32 \times 4}$, i.e., 4096 dimensions), conventional low-dimensional visualization techniques such as 2D PCA were not sufficiently informative in our analyses. To overcome this limitation, we adopt Linear Discriminant Analysis (LDA) to identify the direction that maximizes separation between the two distributions, and visualize the resulting one-dimensional projections as Gaussian-smoothed densities, as shown in Figure 3.

Our results reveal that the distribution gap between generative and encoder latents is smaller for iMF-XL/2 compared to iMF-B/2. This trend is further corroborated by a high-dimensional latent Fréchet distance (Latent-FD, see Appendix A), which yields a lower distance for iMF-XL/2 (91.55) than for iMF-B/2 (103.64), consistent with lower LDA AUC and reduced differences in distributional

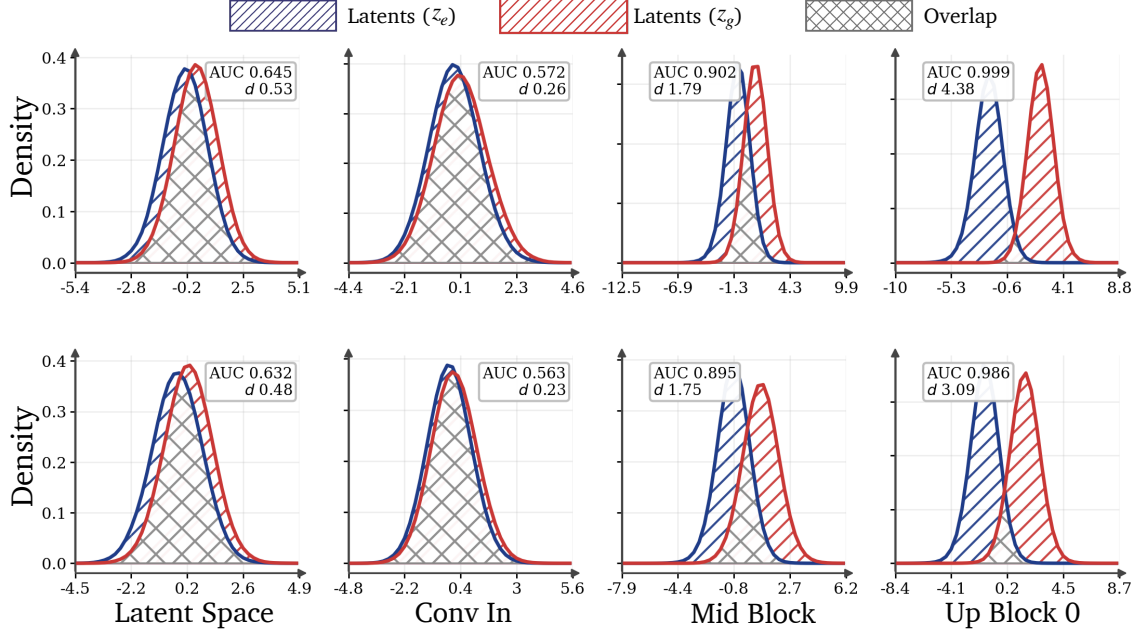


Figure 3: LDA projections of z_e and z_g for iMF-B/2 (top) and iMF-XL/2 (bottom). The vertical axis shows the Gaussian-smoothed, normalized histogram density of the one-dimensional LDA scores. Separation increases from latent space to deeper decoder feature spaces, indicating a progressively amplified distribution mismatch between encoder and generative latents.

means. Together, these results suggest that stronger generative models produce latents that are more closely aligned with the encoder distribution.

Furthermore, we observe a systematic effect when analyzing latents in progressively deeper decoder feature spaces. The separability between the two distributions increases significantly, indicating that the mismatch is amplified as representations propagate through the decoder. At Decoder Up Block 0, the LDA AUC reaches 0.999, demonstrating near-complete separation. We refer to this phenomenon as the *decoder amplification effect*, where small gap in latent space are magnified during decoding. This implies that even relatively mild latent mismatches can lead to pronounced differences in both reconstruction and generative outputs. These observations suggest that latent distribution mismatch is an important factor underlying the gap between reconstruction quality and generative performance.

3.2 Latent Distribution Shift Along the Generative Trajectory

We next analyze the latent behavior of GAR by quantifying the discrepancy between $\mathcal{P}_g^t$ and $\mathcal{P}_e$, as well as between $\mathcal{P}_g^t$ and $\mathcal{P}_g$. Using the SD-VAE encoder [Rom+22] and an iMF-B/2 model [Gen+25b], we extract these latents on ImageNet [Den+09] and compute the corresponding Latent-FD.

As shown in Figure 4, we observe a consistent trend: as the noise level increases, $\mathcal{P}_g^t$ progressively diverges from $\mathcal{P}_e$ while becoming increasingly aligned with $\mathcal{P}_g$. This indicates that $\mathcal{P}_g^t$ forms a trajectory bridging $\mathcal{P}_e$ and $\mathcal{P}_g$, providing a close approximation to the generative process. These results suggest that the gap between reconstruction and generation arises from a systematic distribution shift along the generative trajectory. Standard reconstruction evaluates the decoder under $\mathcal{P}_e$, which differs substantially from the latents encountered

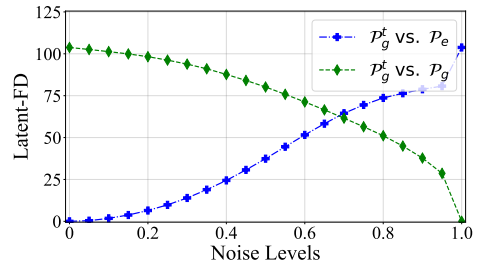


Figure 4: Latent-FD between $\mathcal{P}_g^t$ and $\mathcal{P}_e$, and between $\mathcal{P}_g^t$ and $\mathcal{P}_g$ across noise levels, showing progressive divergence from $\mathcal{P}_e$ and alignment with $\mathcal{P}_g$.

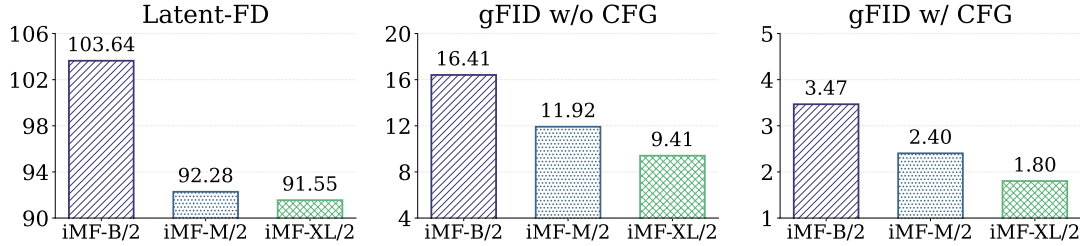


Figure 5: Latent distribution mismatch and generative performance across models. As the Latent-FD between $\mathcal{P}_e$ and $\mathcal{P}_g$ decreases, the corresponding gFID scores (both without and with CFG) also decrease consistently, showing a strong association between latent distribution alignment and generative quality.

during generation, whereas GAR evaluates reconstruction under generation-consistent intermediate latent distributions. This progressive shift is also accompanied by a systematic deterioration in both structural and perceptual reconstruction quality as the noise level increases. A detailed analysis of these effects is provided in [Appendix C](#).

3.3 Linking Latent Distribution Mismatch to Generative Performance

We now investigate how latent distribution mismatch relates to downstream generative performance. Specifically, we quantify the discrepancy between $\mathcal{P}_e$ and $\mathcal{P}_g$ using Latent-FD, and relate it to generation quality measured by FID, both with and without classifier-free guidance (CFG). Our evaluation uses the SD-VAE tokenizer [\[Rom+22\]](#) together with iMF-B/2, iMF-M/2, and iMF-XL/2 [\[Gen+25b\]](#), all trained on the ImageNet dataset [\[Den+09\]](#).

As shown in Figure 5, we observe a clear trend: models with smaller Latent-FD achieve lower gFID, both with and without CFG. This suggests that latent distribution mismatch plays an important role in generative performance. One possible explanation is that a larger mismatch forces the decoder to operate on latents that deviate from its training distribution, leading to degraded sample quality. While not causal, this trend indicates that latent distribution alignment is an important factor. We next provide a theoretical perspective that offers an indirect explanation of this empirical relationship.

Proposition 1 (A Lipschitz Bound Linking Latent-FD and gFID). *Let $\psi : \mathcal{X} \rightarrow \mathbb{R}^{2048}$ denote the Inception feature extractor used for FID, and define the decoder-feature map $T = \psi \circ \mathcal{D}_\phi$. Suppose that $\mathcal{P}_e$ and $\mathcal{P}_g$ are Gaussian latent distributions. Assume that the decoder $\mathcal{D}_\phi$ and feature extractor ψ are Lipschitz with constants L_D and L_ψ , respectively. Then we have*

$$\sqrt{\text{gFID}} \leq \sqrt{\text{rFID}} + L_\psi L_D \sqrt{\text{Latent-FD}(\mathcal{P}_e, \mathcal{P}_g)}. \quad (3.1)$$

The proof is provided in [Appendix B](#). [Proposition 1](#) provides a theoretical lens for the empirical trend in Figure 5. It bounds gFID by rFID and a term proportional to latent distribution mismatch, amplified by the decoder-feature map. This suggests that reducing the discrepancy between $\mathcal{P}_e$ and $\mathcal{P}_g$ improves generative performance.

4 Why rFID Fails to Predict gFID and GAR-FID Tracks It

In this section, we investigate why rFID fails to predict gFID, whereas GAR-FID succeeds. We first use a controlled study with a fixed tokenizer to isolate the effect of the generative model, and then examine whether the resulting explanation generalizes across cross-tokenizer configurations.

4.1 Controlled Study With a Fixed Tokenizer

We begin with a controlled study designed to isolate the role of the generative model. Specifically, we fix the SD-VAE tokenizer [\[Rom+22\]](#) and vary the generative model across three scales: iMF-B/2,

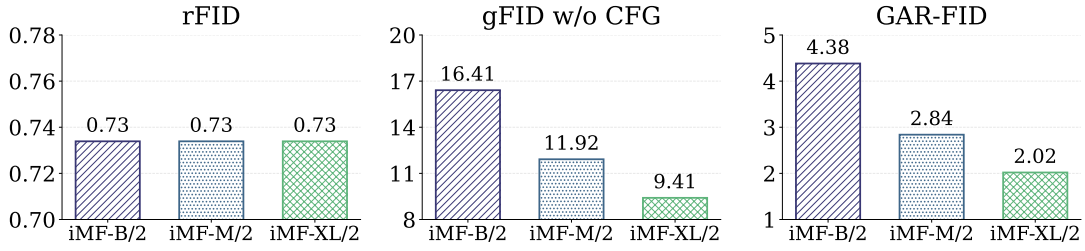


Figure 6: Comparison of rFID, gFID (w/o CFG), and GAR-FID (w/o CFG) under a fixed tokenizer (SD-VAE) as the generative model scale increases (iMF-B/2 $\rightarrow$ iMF-XL/2). While gFID improves with model scale, rFID remains nearly unchanged, whereas GAR-FID follows a similar trend to gFID.

iMF-M/2, and iMF-XL/2 [Gen+25b]. Fixing the tokenizer ensures that all latent distributions are defined in a shared latent space, making Latent-FD($\mathcal{P}_e, \mathcal{P}_g$) well-defined and directly comparable across models. This design allows us to attribute all observed differences to changes in $\mathcal{P}_g$, thereby isolating the effect of latent distribution mismatch on metric behavior.

Figure 6 shows that gFID decreases substantially as model scale increases, indicating improved generative quality. In contrast, rFID remains constant across models, suggesting almost no correlation with generative performance. GAR-FID, however, decreases consistently with gFID, closely tracking improvements in generation. This observation indicates that rFID fails to reflect generative quality, whereas GAR-FID provides a reliable proxy.

Latent Distribution Perspective. To explain this discrepancy, we analyze the relationship between latent distribution mismatch and generative performance. As empirically demonstrated in Section 3.3, we observe a strong empirical association between Latent-FD($\mathcal{P}_e, \mathcal{P}_g$) and gFID, where larger mismatch is consistently associated with worse generative quality. We summarize this relationship as Latent-FD($\mathcal{P}_e, \mathcal{P}_g$) $\uparrow \rightsquigarrow$ gFID $\uparrow$, and refer to it as an *empirical proxy relationship*, indicating that Latent-FD($\mathcal{P}_e, \mathcal{P}_g$) serves as a useful empirical proxy for generative performance in this setting. This suggests that generative quality is strongly associated with the discrepancy between the encoder-induced distribution $\mathcal{P}_e$ and the generative distribution $\mathcal{P}_g$.

Why rFID Fails to Predict gFID. The failure of rFID stems from a mismatch between its evaluation protocol and the generative process. Specifically, rFID evaluates reconstruction under encoder latents $z_e \sim \mathcal{P}_e$ and is therefore insensitive to the discrepancy between $\mathcal{P}_e$ and $\mathcal{P}_g$. As a result, it cannot capture Latent-FD($\mathcal{P}_e, \mathcal{P}_g$) and fails to serve as a reliable proxy for generative performance.

Why GAR-FID Predicts gFID. In contrast, GAR-FID evaluates reconstruction under intermediate generative latents $z_g^t \sim \mathcal{P}_g^t$, which better reflect the distributions encountered during generation. As shown in Figure 7, GAR-FID exhibits near-perfect rank correlation with Latent-FD($\mathcal{P}_g^t, \mathcal{P}_e$) across the evaluated noise levels (Spearman’s $\rho = 1.0$, $p < 10^{-8}$), indicating that it effectively captures the mismatch between intermediate generative latents and the encoder-induced distribution. This consistent empirical relationship suggests that larger intermediate latent mismatch is associated with worse GAR-FID, which we summarize as GAR-FID $\uparrow \rightsquigarrow$ Latent-FD($\mathcal{P}_g^t, \mathcal{P}_e$) $\uparrow$.

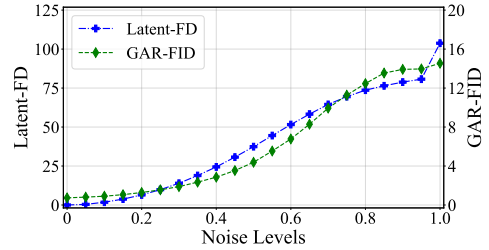


Figure 7: GAR-FID (w/o CFG) closely tracks Latent-FD($\mathcal{P}_g^t, \mathcal{P}_e$) across noise levels.

Moreover, under the fixed-tokenizer setting, Latent-FD($\mathcal{P}_g^t, \mathcal{P}_e$) preserves the model ordering of Latent-FD($\mathcal{P}_g, \mathcal{P}_e$) across noise levels (Appendix D), suggesting that intermediate latent mismatch can serve as an informative empirical proxy for the final latent mismatch, which preserves model ordering and links GAR-FID to final generative performance. We summarize this relationship as Latent-FD($\mathcal{P}_g^t, \mathcal{P}_e$) $\uparrow \rightsquigarrow$ Latent-FD($\mathcal{P}_g, \mathcal{P}_e$) $\uparrow$.

Table 1: Correlation between FID-based metrics and generative performance (gFID). Correlations are computed across models obtained by training SiT with 13 pre-trained VAEs. We report both Pearson correlation (PCC) and Spearman rank correlation (SRCC) under different settings (with and without CFG) for SiT-B, SiT-XL, and their mixed configurations. GAR-FID without CFG consistently exhibits stronger correlation with gFID across noise levels η_t , whereas the standard reconstruction metric (rFID) shows negative correlation. **Bold** indicates the best results. [†] denotes results from [Xu+26], and * indicates our reproduced results. The full results are provided in Table 7 in Appendix G.

Metrics	gFID SiT/B				gFID SiT/XL				gFID SiT/mixed			
	w/o CFG		w/ CFG		w/o CFG		w/ CFG		w/o CFG		w/ CFG	
	PCC (†)	SRCC (†)	PCC (†)	SRCC (†)	PCC (†)	SRCC (†)	PCC (†)	SRCC (†)	PCC (†)	SRCC (†)	PCC (†)	SRCC (†)
rFID [†]	-0.04	-0.31	-0.07	-0.31	-0.06	-0.21	-0.15	-0.31	–	–	–	–
iFID [†]	0.85	0.86	0.82	0.84	0.89	0.91	0.88	0.92	–	–	–	–
rFID*	-0.11	-0.36	-0.11	-0.30	-0.13	-0.34	-0.15	-0.39	-0.10	-0.30	-0.10	-0.28
iFID*	0.86	0.83	0.80	0.79	0.90	0.88	0.88	0.90	0.72	0.77	0.64	0.70
GAR-FID ($\eta_t = 0.6$)	0.81	0.79	0.77	0.78	0.79	0.84	0.80	0.79	0.86	0.89	0.84	0.85
GAR-FID ($\eta_t = 0.7$)	0.93	0.90	0.89	0.87	0.91	0.90	0.91	0.86	0.95	0.96	0.93	0.91
GAR-FID ($\eta_t = 0.8$)	0.98	0.96	0.96	0.94	0.98	0.99	0.97	0.96	0.99	0.99	0.97	0.95
GAR-FID ($\eta_t = 0.9$)	1.00	0.99	0.98	0.98	1.00	1.00	0.98	0.96	1.00	1.00	0.98	0.97
GAR-FID ($\eta_t = 1.0$)	1.00	0.99	0.98	0.99	1.00	1.00	0.98	0.96	1.00	1.00	0.98	0.97

Taken together with the empirical proxy relationship between Latent-FD($\mathcal{P}_e, \mathcal{P}_g$) and gFID demonstrated in Section 3.3, these observations provide a plausible explanation for why GAR-FID tracks gFID in the controlled setting. Empirically, our results suggest the following association chain:

$$\text{GAR-FID} \uparrow \rightsquigarrow \text{Latent-FD}(\mathcal{P}_g^t, \mathcal{P}_e) \uparrow \rightsquigarrow \text{Latent-FD}(\mathcal{P}_g, \mathcal{P}_e) \uparrow \rightsquigarrow \text{gFID} \uparrow, \quad (4.1)$$

supporting the use of GAR-FID as a useful empirical proxy for generative performance.

Scope of the Controlled Setting. We emphasize that this analysis is conducted under a fixed tokenizer setting. Varying the tokenizer would change the underlying latent space, making Latent-FD($\mathcal{P}_e, \mathcal{P}_g$) not directly comparable across configurations. Our goal here is to isolate the effect of the generative model and characterize a systematic relationship between latent distribution mismatch and metric behavior.

4.2 Generalization Across Tokenizers and Generative Models

We next examine whether the predictive behavior observed in the controlled setting generalizes when both the tokenizer and the generative model vary. Following iFID [Xu+26], we evaluate 13 tokenizers with two latent generative models, SiT-B and SiT-XL [Ma+24], and compute rFID, GAR-FID, and gFID for each tokenizer–generator configuration. All computed metrics are reported in Tables 4 and 5, with detailed experimental settings provided in Appendix G.

As shown in Table 1, rFID exhibits consistently negative correlation with gFID, both with and without CFG, across SiT-B, SiT-XL, and their mixed configurations. This indicates that better reconstruction performance can correspond to worse generative performance, consistent with the reconstruction–generation dilemma observed in prior work [YW25; Kou+25; Ye+25; Sko+25; Che+25b; Xu+26]. In contrast, GAR-FID without CFG shows increasingly strong positive correlation with gFID as the noise level η_t increases. When $\eta_t \geq 0.7$, GAR-FID outperforms iFID [Xu+26] on both SiT-B and SiT-XL; when $\eta_t \geq 0.8$, it achieves near-perfect correlation with gFID across most settings.

The mixed SiT-B/XL setting further tests whether this correlation is preserved across generator scales. Correlations computed within a single generative model, such as SiT-B or SiT-XL [Ma+24], primarily measure how a metric responds to tokenizer variation under a fixed generator scale. By contrast, the mixed setting evaluates whether the metric remains predictive when tokenizer choice and generator capacity vary jointly, which better reflects cross-system comparisons in latent visual generative modeling. In this more demanding regime, iFID shows weaker correlation with gFID, whereas GAR-FID exhibits slightly stronger correlation, indicating better robustness across tokenizer and generator variations. These results suggest that the predictive behavior identified in the controlled setting extends to more realistic cross-tokenizer and cross-generator comparisons, providing strong empirical evidence that GAR-FID is a robust proxy for generative performance.

5 Latent Distribution Alignment as a Unifying Perspective

As discussed in Section 4, GAR reveals that the gap between reconstruction and generation arises from evaluating the decoder under mismatched latent distributions. This observation suggests that improving generative performance can be viewed as a problem of aligning training with the latent distributions encountered at evaluation time. From this perspective, GAR is not merely an evaluation metric, but exposes a deeper insight: *evaluation and training should be unified under generative or intermediate generative latent distributions*.

This perspective naturally leads to two complementary ways to achieve such alignment. The first is to improve the latent generative process, such that the generated distribution $\mathcal{P}_g$ better matches the encoder distribution $\mathcal{P}_e$. The second is to adapt the decoder, such that it reliably operates on samples drawn from $\mathcal{P}_g$ or intermediate generative distributions $\mathcal{P}_g^t$. These correspond to reducing distribution mismatch on the generator side, or aligning the decoder with the generative latent distribution.

A large body of prior work has focused on the first direction, improving the generative distribution $\mathcal{P}_g$ through multiple complementary avenues, including stronger architectures and scaling strategies, improved objectives and transport formulations, enhanced multimodal conditioning, and efficient sampling or distillation techniques [PX22; Ess+24; Mei+24; Ton+26; Che+24; Lip+23; LGL23; AVE23; Gen+25a; Wu+25; Lab24; Xie+25; SH22; Son+23; Luo+23; SD24; Sau+24]. Additional approaches leverage reinforcement learning, preference-based optimization, and richer latent representations to further improve generative quality [Yu+25; Liu+25; Xue+25; Lee+23; Fan+23; Xu+23; Wal+23; Zhe+26; Len+25; Yan+26; Che+25b; Che+25c; Ye+25]. Despite their diversity, these approaches share a common goal: reshaping $\mathcal{P}_g$ so that generated latents lie in regions of the latent space that can be more reliably decoded. From the perspective of GAR, they can be viewed as implicitly improving the compatibility between $\mathcal{P}_g$ and the decoder.

However, these approaches address only one side of the problem and overlook a fundamental inconsistency: the decoder is trained under $\mathcal{P}_e$, but evaluated under $\mathcal{P}_g$. This training–evaluation mismatch is inherent in many latent generative models and is unlikely to be resolved by improving the generator alone. Moreover, as evidenced by the *decoder amplification effect*, even small discrepancies between $\mathcal{P}_e$ and $\mathcal{P}_g$ can be significantly amplified during decoding, leading to pronounced degradation in both reconstruction and generative outputs. Consequently, generative performance depends on how well $\mathcal{D}_\phi$ operates on $\mathcal{P}_g$, while the decoder itself is never optimized for this distribution.

From the perspective of evaluation–training unification, this implies that the decoder should be optimized under the same latent distribution on which it is evaluated. In contrast to standard reconstruction training, which operates on $z_e \sim \mathcal{P}_e$, this requires aligning decoder training with generation-consistent intermediate latents. We instantiate this idea through *decoder adaptation* (Section 6), a simple procedure that fine-tunes the decoder on intermediate generative latents. By explicitly aligning decoder training with the intermediate generative latent distribution, this approach provides a natural and promising route to improving both generation-aware reconstruction and downstream generative performance.

6 Decoder Adaptation via Latent Distribution Alignment

We operationalize this idea of latent distribution alignment via *decoder adaptation*, a simple yet effective procedure for aligning the decoder with the latent distribution encountered during generation. As described in Section 2, the encoder $\mathcal{E}_\theta$ maps an image x to a latent representation $z_e = \mathcal{E}_\theta(x)$, from which a noisy latent is constructed as $z_{\text{noisy}}^t = a_t z_e + b_t \varepsilon$. The generative model $\mathcal{G}_\varphi$ then produces intermediate generative latents $z_g^t = \mathcal{G}_\varphi(z_{\text{noisy}}^t) \sim \mathcal{P}_g^t$, which are decoded as $\bar{x}(\phi) = \mathcal{D}_\phi(z_g^t)$.

Unlike standard reconstruction, where the decoder is trained on encoder latents $z_e \sim \mathcal{P}_e$, our approach explicitly optimizes the decoder on intermediate generative latents $z_g^t \sim \mathcal{P}_g^t$. From the

Table 2: Effect of decoder adaptation (DA) on generative performance. We report gFID without and with CFG across iMF model scales, comparing results before and after DA under different evaluation protocols. DA consistently improves performance across all settings. [†]: results reported in iMF [Gen+25b]. [‡]: our reproduction using the official iMF protocol. *: our reproduction using the OpenAI protocol. See Appendix I for protocol details.

Setting	gFID w/o CFG (↓)				gFID w/ CFG (↓)			
	iMF-B/2	iMF-M/2	iMF-L/2	iMF-XL/2	iMF-B/2	iMF-M/2	iMF-L/2	iMF-XL/2
w/o DA [†]	-	-	-	-	3.39	2.27	1.86	1.72
w/o DA [‡]	16.58	11.92	9.28	9.78	3.37	2.27	1.86	1.73
w/ DA [‡]	12.21 ↑4.37	9.52 ↑2.40	7.34 ↑1.94	7.47 ↑2.31	2.90 ↑0.47	2.12 ↑0.15	1.65 ↑0.21	1.56 ↑0.17
w/o DA*	16.41	11.92	9.26	9.41	3.47	2.40	1.90	1.80
w/ DA*	12.41 ↑4.00	9.66 ↑2.26	7.41 ↑1.85	7.61 ↑1.80	3.12 ↑0.35	2.30 ↑0.10	1.80 ↑0.10	1.73 ↑0.07

perspective of evaluation–training unification, this ensures that the decoder is trained under a latent distribution closer to that encountered at evaluation time. As generative performance depends on how well the decoder operates on latents drawn from $\mathcal{P}_g$, aligning training with $\mathcal{P}_g^t$ provides a practical way to improve both generation-aware reconstruction and downstream generative quality.

To train the decoder, we adopt a composite objective commonly used in modern visual tokenizers [Tia+24; Fan+26; Yu+26], combining pixel-level, perceptual, and adversarial supervision:

$$\mathcal{L}(\phi) = \lambda_1 \|\bar{x}(\phi) - x\|_2^2 + \lambda_2 \mathcal{L}_{\text{LPIPS}} + \lambda_3 \mathcal{L}_{\text{GAN}} + \lambda_4 \mathcal{L}_{\text{Gram}}. \quad (6.1)$$

Here, all losses are computed between the reconstruction $\bar{x}(\phi)$ and the original input x . $\mathcal{L}_{\text{LPIPS}}$ measures perceptual similarity using deep feature representations [Zha+18; SZ15], $\mathcal{L}_{\text{GAN}}$ is implemented as a hinge-based adversarial loss [Iso+17; LY17; KLA19; Kar+20], and $\mathcal{L}_{\text{Gram}}$ matches feature covariance statistics via Gram matrices [GEB16; Lu+25]. While these components are standard, their role here is essential for stable training: they enable robust and high-fidelity decoding under the shifted latent distribution $\mathcal{P}_g^t$. We set $\lambda_1 = 1.0$, $\lambda_2 = 1.0$, $\lambda_3 = 0.7$, and $\lambda_4 = 10.0$ in all experiments. For adversarial training, we follow recent advances in visual tokenizer learning [Tia+24; Che+25c; Fan+26; Li+25], and employ a frozen DINO-S discriminator [Car+21; Oqu+24], which provides strong semantic supervision. To further stabilize training, we incorporate differentiable augmentations (DiffAug) [Zha+20], consistency regularization [Zha+19], and LeCAM regularization [Tse+21]. Further details on noise-level selection are provided in Appendix E.

7 Experiments

Experimental Setup. To evaluate the effectiveness of decoder adaptation, we conduct experiments on a family of latent generative models based on iMF [Gen+25b], all of which share a common tokenizer, SD-VAE [Rom+22]. We consider four model scales: iMF-B/2, iMF-M/2, iMF-L/2, and iMF-XL/2. During adaptation, both the tokenizer encoder and the latent generative model are kept frozen, and only the tokenizer decoder parameters are updated. The generative model is applied without CFG throughout the adaptation training process. Experiments are conducted on ImageNet-1k at 256×256 resolution [Den+09]. Additional implementation details are provided in Appendix J. Code and pretrained checkpoints will be publicly released to ensure reproducibility.

Main Results. As shown in Table 2, decoder adaptation consistently improves generative performance across all settings, with larger gains observed without CFG and for smaller-scale models (e.g., iMF-B/2). Importantly, it introduces no additional inference cost, as it only updates the decoder parameters without altering the tokenizer or generative model. Additional results are provided in Appendix K, including reconstruction performance (Table 10) and comparisons with additional baselines (Table 11). Visually, decoder adaptation reduces artifacts and yields more semantically consistent samples (Figures 13–23).

Decoder Adaptation Makes rFID Predictive. Figure 8 shows the relationship between rFID and gFID (w/o CFG) across model scales, before and after decoder adaptation. Before adaptation, rFID

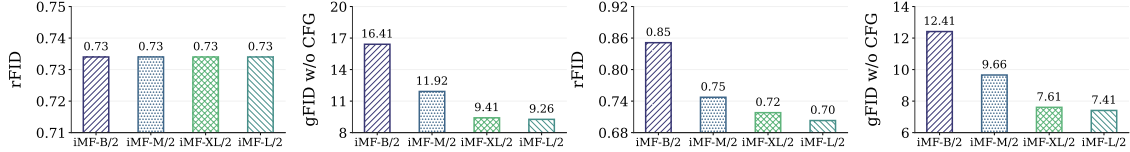


Figure 8: rFID vs. gFID (w/o CFG) across model scales, before (left two) and after (right two) decoder adaptation. rFID is uninformative before adaptation but becomes aligned with gFID after adaptation.

Table 3: Effect of fixed noise levels in decoder adaptation. We report gFID without CFG, evaluated using the OpenAI protocol, and rFID across iMF model scales for decoder adaptation with different fixed noise levels η_t . See Appendix I for protocol details.

Noise Level	gFID w/o CFG (↓)				rFID (↓)			
	iMF-B/2	iMF-M/2	iMF-L/2	iMF-XL/2	iMF-B/2	iMF-M/2	iMF-L/2	iMF-XL/2
$\eta_t = 0.0^\dagger$	16.41	11.92	9.26	9.41	0.73	0.73	0.73	0.73
$\eta_t = 0.0$	15.27	11.30	8.38	8.82	0.58	0.62	0.59	0.62
$\eta_t = 0.1$	14.76	11.03	8.19	8.64	0.62	0.61	0.58	0.60
$\eta_t = 0.2$	13.50	10.15	7.89	8.03	0.71	0.67	0.62	0.64
$\eta_t = 0.3$	12.86	9.99	7.57	7.84	0.77	0.70	0.67	0.67
$\eta_t = 0.4$	12.08	9.52	7.51	7.24	0.94	0.79	0.75	0.74
$\eta_t = 0.5$	11.58	9.18	7.36	7.19	1.20	1.01	0.94	0.89
$\eta_t = 0.6$	11.88	9.34	7.41	7.40	1.81	1.42	1.26	1.21

remains constant across models and fails to reflect variations in gFID. After adaptation, rFID decreases consistently with gFID, recovering its predictive ability. This suggests that the lack of predictiveness of rFID arises from a mismatch between the decoder and the generative latent distribution. By adapting the decoder to generation-consistent latents, this mismatch is mitigated, enabling rFID to better reflect generative performance.

Sensitivity Analyses on Noise Levels. To study the effect of the noise level used for decoder adaptation, we conduct a sensitivity analysis by fixing $\eta_t \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6\}$ during adaptation. As shown in Table 3, $\eta_t = 0.5$ achieves the best gFID without CFG across all iMF model scales. When η_t is too small, the intermediate generative latents $z_g^t \sim \mathcal{P}_g^t$ remain insufficiently aligned with the true generative latent distribution $\mathcal{P}_g$, resulting in smaller gains in gFID. Increasing η_t exposes the decoder to more generation-like latents, but overly large noise levels can make the paired reconstruction objective between $\bar{x}$ and x less well posed. For example, at $\eta_t = 0.6$, $z_g^t \sim \mathcal{P}_g^t$ may deviate substantially from the encoder-induced distribution $\mathcal{P}_e$, making naive reconstruction alignment more prone to degenerate behavior. Consistently, rFID increases monotonically with η_t across all iMF model scales. These results indicate that the noise level for decoder adaptation should be carefully chosen to balance generative performance and reconstruction fidelity.

8 Conclusion

We showed that the gap between reconstruction quality and generative performance stems from an evaluation misalignment: reconstruction evaluates the decoder on encoder latents, whereas generation relies on a different latent distribution. Based on this observation, we introduced generation-aware reconstruction (GAR), which evaluates the decoder on intermediate generative latents and produces a metric, GAR-FID, that strongly tracks gFID in practice. We further proposed decoder adaptation to align decoder training with intermediate generative latents, leading to improved generative quality. These results suggest latent distribution alignment provides a unifying perspective for both evaluation and optimization in latent generative models. Our framework improves the fidelity of evaluating and optimizing generative models, which may indirectly lead to stronger generative capabilities and potential misuse (e.g., generating misleading or harmful content). While it does not introduce new generative architectures, it provides a more reliable evaluation framework that may help identify and mitigate failure modes. We encourage responsible use and further research on safeguards.

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Appendix

A Latent Fréchet Distance

To measure the distributional discrepancy in latent space, we adopt the Fréchet distance between two sets of latent variables, $z_i^a \sim \mathcal{P}_A$ and $z_i^b \sim \mathcal{P}_B$. Specifically, we estimate the empirical mean and covariance of the two latent distributions, denoted by (μ_1, Σ_1) and (μ_2, Σ_2) , respectively. Based on these statistics, we define the latent Fréchet distance (Latent-FD) as

$$\text{FD}(\mathcal{P}_A, \mathcal{P}_B) = \|\mu_1 - \mu_2\|_2^2 + \text{Tr}(\Sigma_1 + \Sigma_2 - 2(\Sigma_1 \Sigma_2)^{1/2}), \quad (\text{A.1})$$

This metric provides a well-established measure of discrepancy between two distributions by capturing both first-order (mean) and second-order (covariance) statistics, and is closely related to the widely used Fréchet Inception Distance (FID). Both metrics compute the Fréchet distance under a Gaussian assumption, differing only in the representation space used to estimate these statistics. Specifically, FID operates in a learned semantic feature space, where each image is mapped to a 2048-dimensional representation using a pretrained InceptionV3 [Sze+16] network. In contrast, our Latent-FD is computed directly in the model latent space, using latents from the encoder or generative process. This distinction allows Latent-FD to directly quantify distributional discrepancies relevant to the generative pipeline, while retaining the same underlying geometric interpretation as FID.

B Proof of Proposition 1

Let $\mathcal{P}_{\text{data}}$ denote the real image distribution, let ψ be the Inception feature extractor used for FID, and define $T = \psi \circ \mathcal{D}_\phi$. We write

$$\mathcal{Q}_{\text{data}} = \psi_{\#} \mathcal{P}_{\text{data}}, \quad \mathcal{Q}_e = T_{\#} \mathcal{P}_e, \quad \mathcal{Q}_g = T_{\#} \mathcal{P}_g.$$

For any distribution $\mathcal{Q}$ with finite second moment, let $\Gamma(\mathcal{Q}) = \mathcal{N}(\mu_{\mathcal{Q}}, \Sigma_{\mathcal{Q}})$ be the Gaussian distribution with the same mean and covariance as $\mathcal{Q}$. The FID between two feature distributions $\mathcal{Q}$ and $\mathcal{R}$ is the squared 2-Wasserstein distance between their Gaussian approximations:

$$\text{FID}(\mathcal{Q}, \mathcal{R}) = W_2^2(\Gamma(\mathcal{Q}), \Gamma(\mathcal{R})).$$

Therefore, using the triangle inequality for W_2 ,

$$\begin{aligned} \sqrt{\text{gFID}} &= W_2(\Gamma(\mathcal{Q}_{\text{data}}), \Gamma(\mathcal{Q}_g)) \\ &\leq W_2(\Gamma(\mathcal{Q}_{\text{data}}), \Gamma(\mathcal{Q}_e)) + W_2(\Gamma(\mathcal{Q}_e), \Gamma(\mathcal{Q}_g)) \\ &= \sqrt{\text{rFID}} + \sqrt{\text{FID}(\mathcal{Q}_e, \mathcal{Q}_g)}. \end{aligned}$$

The Gaussian Fréchet distance between the moment-matched Gaussians is upper bounded by the true Wasserstein distance between the original distributions, since

$$\text{FID}(\mathcal{Q}_e, \mathcal{Q}_g) \leq W_2^2(\mathcal{Q}_e, \mathcal{Q}_g).$$

Moreover, because T is L -Lipschitz, pushing any coupling of $\mathcal{P}_e$ and $\mathcal{P}_g$ through T gives

$$W_2(\mathcal{Q}_e, \mathcal{Q}_g) = W_2(T_{\#} \mathcal{P}_e, T_{\#} \mathcal{P}_g) \leq L W_2(\mathcal{P}_e, \mathcal{P}_g).$$

Finally, since $\mathcal{P}_e$ and $\mathcal{P}_g$ are Gaussian, the latent Fréchet distance equals the squared Wasserstein distance between them:

$$\text{Latent-FD}(\mathcal{P}_e, \mathcal{P}_g) = W_2^2(\mathcal{P}_e, \mathcal{P}_g).$$

Combining the preceding inequalities yields

$$\sqrt{\text{gFID}} \leq \sqrt{\text{rFID}} + L_{\psi} L_{\mathcal{D}} \sqrt{\text{Latent-FD}(\mathcal{P}_e, \mathcal{P}_g)}.$$



Figure 9: Illustration of *structural collapse* under increasing noise levels in GAR reconstruction. Each row shows an original input image (leftmost column) followed by GAR reconstructions at noise levels $\eta_t \in \{0.0, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0\}$ from left to right. As the noise level increases, reconstructions progressively deviate from the original inputs: while coarse semantic content may remain loosely recognizable at intermediate noise levels, fine-grained structural details and local textures are gradually degraded. At high noise levels (e.g., $\eta_t \geq 0.6$), the reconstructed images no longer preserve the original structure, resulting in semantically plausible but structurally inconsistent outputs. This phenomenon highlights the breakdown of structural fidelity along the generative trajectory.

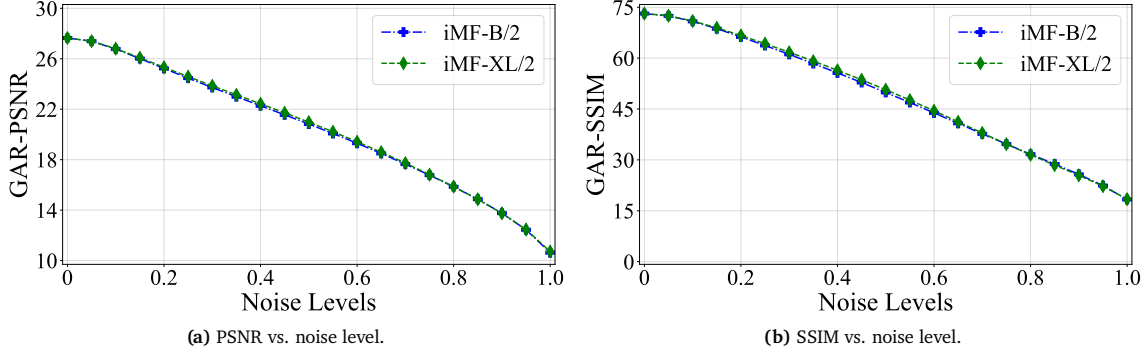


Figure 10: Structural quality under GAR across noise levels. PSNR (left) and SSIM (right) degrade consistently as noise increases. Increasing generative model capacity (iMF-B/2 vs. iMF-XL/2) does not improve structural quality, as both follow nearly identical trends.

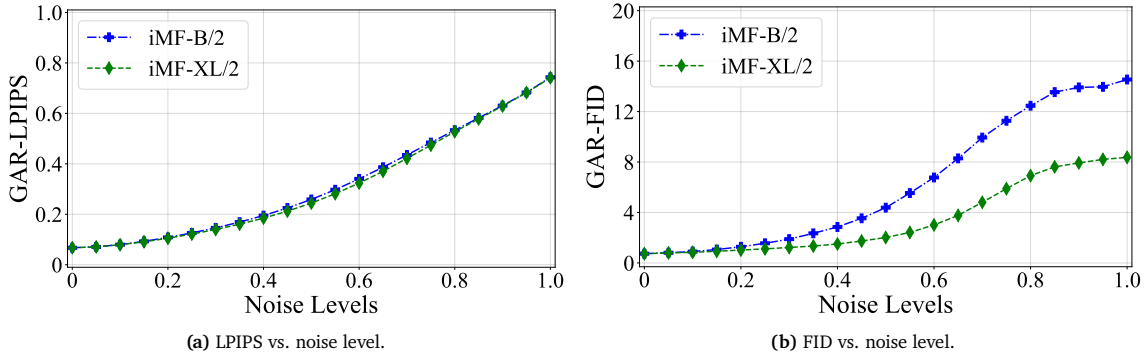


Figure 11: Perceptual quality under GAR across noise levels. LPIPS (left) and FID without CFG (right) both degrade as noise increases. Increasing model capacity (iMF-B/2 vs. iMF-XL/2) has little effect on LPIPS but significantly improves FID across noise levels.

C Structural and Perceptual Quality under GAR

We provide additional analysis of reconstruction quality under GAR using both structural and perceptual metrics. This analysis complements the main text by illustrating how latent distribution shift manifests in reconstruction quality. For structural fidelity, we consider peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). For perceptual quality, we use Fréchet Inception Distance [FID; Heu+17] and learned perceptual image patch similarity [LPIPS; Zha+18]. When computed under the GAR framework, we refer to these metrics as

GAR-PSNR, GAR-SSIM, GAR-FID, and GAR-LPIPS, respectively. All experiments are conducted on the ImageNet validation set [Den+09], using the SD-VAE tokenizer [Rom+22] and two representative flow-matching models, iMF-B/2 and iMF-XL/2 [Gen+25b].

Structural Quality. As depicted in Figure 10, both structural quality metrics, PSNR and SSIM, exhibit a consistent decline as the noise level increases. Notably, when the noise level becomes sufficiently large (e.g., $\eta_t \geq 0.6$), the GAR-reconstructed images diverge substantially from the original inputs. While coarse global semantic structure may still be partially preserved, fine-grained local textures are largely degraded. We refer to this phenomenon as *structural collapse*, as illustrated in Figure 9. Another notable observation is that increasing generative model capacity (iMF-B/2 vs. iMF-XL/2) does not lead to improvements in structural quality, as both models exhibit nearly identical degradation trends across noise levels. This suggests that stronger generative models may not substantially improve structural quality in this setting.

Perceptual Quality. As shown in Figure 11, both perceptual metrics, LPIPS and FID without CFG, increase consistently as the noise level grows (higher values indicate worse quality). Notably, increasing generative model capacity (iMF-B/2 vs. iMF-XL/2) has little effect on LPIPS, as the curves nearly overlap, but leads to a substantial improvement in FID across all noise levels. This indicates that while stronger generative models do not improve input-level perceptual similarity, they are associated with improved distribution-level perceptual realism.

Latent Distribution Shift Perspective. We observe that as the noise level increases, both structural and perceptual quality degrade substantially. This behavior is consistent with the latent distribution shift observed along the generative trajectory. As shown in Figure 4, the discrepancy between $\mathcal{P}_g^t$ and $\mathcal{P}_e$ increases monotonically with the noise level, indicating that the latents encountered during decoding progressively move away from the encoder distribution on which the decoder is trained. This leads the decoder to operate under an out-of-distribution regime, which is associated with a consistent degradation in reconstruction quality. This growing mismatch manifests as a loss of structural fidelity and a decline in perceptual quality, as evidenced by the trends in PSNR/SSIM and LPIPS/FID. These findings suggest that such degradation is closely associated with the increasing distributional shift along the generative trajectory.

D Intermediate Latent-FD as a Proxy for Final Latent Mismatch

Figure 12 compares Latent-FD($\mathcal{P}_g^t, \mathcal{P}_e$) across noise levels for different generative models, where $\eta_t = 1.0$ corresponds to Latent-FD($\mathcal{P}_g, \mathcal{P}_e$). We observe that the model ordering is preserved across all evaluated noise levels. This consistent ordering indicates that larger intermediate latent mismatch is associated with larger final mismatch at $\eta_t = 1.0$. We summarize this empirical relationship as

$$\text{Latent-FD}(\mathcal{P}_g^t, \mathcal{P}_e) \uparrow \rightsquigarrow \text{Latent-FD}(\mathcal{P}_g, \mathcal{P}_e) \uparrow.$$

Under the fixed-tokenizer setting, this suggests that Latent-FD($\mathcal{P}_g^t, \mathcal{P}_e$) can serve as a useful proxy for Latent-FD($\mathcal{P}_g, \mathcal{P}_e$) along the generative trajectory.

E Noise Level Choice in Decoder Adaptation

A key practical consideration in decoder adaptation is the choice of noise level. In principle, generation-consistent intermediate latents can be constructed using noise levels $\eta_t \in (0, 1)$. However, the choice of η_t critically affects the validity of the reconstruction objective. When the noise level is too large, the resulting latents undergo severe structural corruption, leading to reconstructions $\bar{x}$ that no longer share semantic content with the original input x . In this regime, applying reconstruction losses between $\bar{x}$ and x becomes ill-posed, and naively aligning them can lead to unstable or degenerate optimization. On the other hand, when the noise level is too small, $\bar{x}$ remains overly similar to x , resulting in a weak training signal that limits effective decoder adaptation under $\mathcal{L}(\phi)$. To balance these two regimes, we restrict the noise level to an intermediate range and sample $\eta_t \sim \text{Unif}[\eta_{\min}, \eta_{\max}]$. In all experiments, we set $\eta_{\min} = 0.25$ and $\eta_{\max} = 0.45$, which we find to provide a stable and informative training signal in practice.

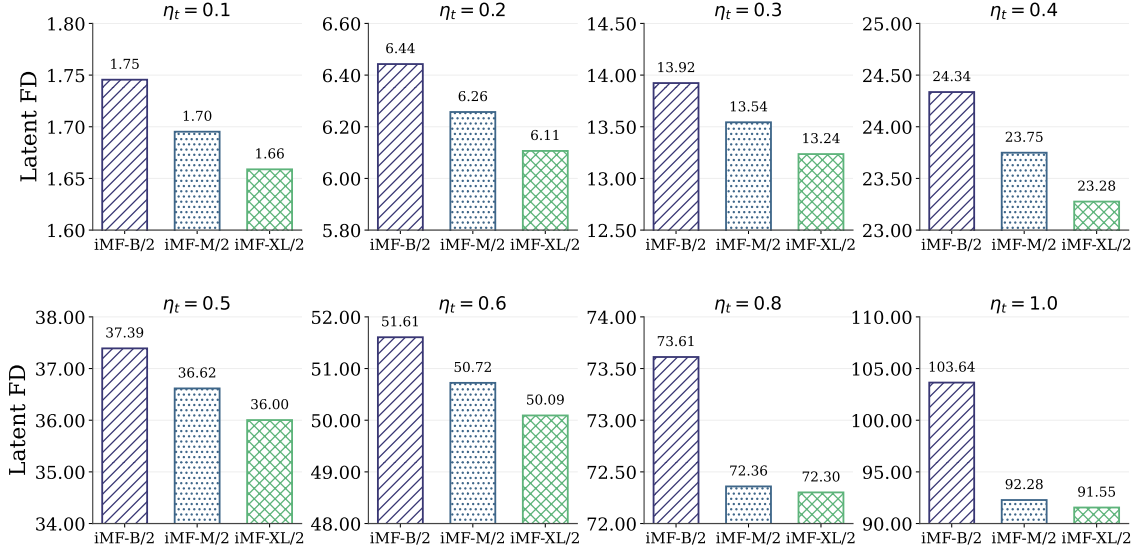


Figure 12: $\text{Latent-FD}(\mathcal{P}_g^t, \mathcal{P}_e)$ across noise levels for different generative models, with $\eta_t = 1.0$ corresponding to $\text{Latent-FD}(\mathcal{P}_g, \mathcal{P}_e)$. The model ordering is preserved across noise levels.

F Limitations

Our work has several limitations.

First, our study focuses on latent generative models based on diffusion and flow-matching paradigms. While the notion of latent distribution mismatch may extend more broadly, we do not investigate its applicability to autoregressive latent generators or other generation frameworks.

Second, our explanation for why GAR-FID predicts gFID is primarily empirical. Although we observe a consistent association between GAR-FID, intermediate latent mismatch, final latent mismatch, and generative performance, we do not provide a formal theoretical guarantee for these relationships.

Third, our latent distribution analysis relies on comparisons within a shared latent space. In particular, the connection between Latent-FD and gFID is most directly observed in a fixed-tokenizer setting with a limited number of comparable model instances, due to the scarcity of publicly available checkpoints. While we complement this with broader cross-tokenizer experiments, distribution-level comparisons across different tokenizers remain challenging.

Finally, GAR and decoder adaptation introduce additional computational overhead, as they require passing latents through the generative model during evaluation or training. Moreover, their performance can depend on the choice of noise level, suggesting that further work is needed to better understand trajectory parameterization and to better understand how to select these parameters.

In addition, GAR-FID remains a proxy for generative performance rather than a direct substitute for gFID. Although it provides a substantially more predictive evaluation signal than standard reconstruction metrics, it does not eliminate the need for direct generative evaluation when final sample quality is the primary target.

G Additional Results for Cross-Tokenizer and Cross-Model Generalization

To examine whether the predictive behavior observed in the controlled setting (Section 4.1) generalizes when both the tokenizer and the generative model vary, we follow iFID [Xu+26] and consider 13 VAEs with publicly available checkpoints. The details of these models are provided in Table 6. These VAEs cover a diverse set of design choices. Some models, such as SD-VAE [Rom+22], IN-VAE [Len+25], QW-VAE [Wu+25], FLUX-VAE [Lab24], and SD3-VAE [Ess+24], are optimized primarily for reconstruction. Others incorporate additional inductive biases, such as equivariance regularization in EQ-VAE [Kou+25]. A third group leverages contrastive learning-based image encoders, including VA-VAE [YW25], SOFT-VQ [Che+25c], MAE-TOK [Che+25b], DE-TOK [Che+25b], DM-VAE [Ye+25], REPAE-VAE [Len+25], and RAE [Zhe+26].

Table 4: Quantitative results for 13 tokenizers. For each tokenizer, we evaluate rFID, gFID (w/o and w/ CFG), and GAR-FID without CFG at noise levels η_t ranging from 0.1 to 1.0. All results are obtained using SiT/B models trained on the corresponding tokenizer [Xu+26].

Tokenizers	rFID ($\downarrow$)	gFID ($\downarrow$)		GAR-FID (η_t) ($\downarrow$)									
		w/o CFG	w CFG	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
SD-VAE [Rom+22]	0.73	46.64	9.91	1.16	2.28	4.85	9.63	17.10	25.94	33.98	38.91	41.35	42.03
IN-VAE [Len+25]	0.26	48.28	10.87	0.46	0.88	1.86	3.86	7.68	14.53	24.37	35.13	41.78	43.36
FLUX-VAE [Lab24]	0.16	62.92	14.38	1.05	2.32	5.42	11.09	18.97	29.14	39.71	49.25	55.71	58.33
QwenImage-VAE [Wu+25]	1.52	48.03	10.97	2.14	2.94	4.54	7.14	11.17	16.84	24.20	32.57	40.49	43.72
SD3-VAE [Ess+24]	0.21	51.58	11.89	0.59	1.06	2.25	4.76	9.34	16.36	26.03	35.92	43.70	46.82
EQ-VAE [Kou+25]	0.59	37.49	9.32	3.37	6.54	10.11	13.91	17.79	21.83	26.12	29.70	32.06	33.26
VA-VAE-16 [YW25]	0.30	18.88	6.01	2.52	3.71	4.91	6.21	7.46	8.85	10.48	12.57	14.90	15.78
VA-VAE-64 [YW25]	0.15	33.56	7.76	1.90	3.10	5.02	7.44	10.32	13.48	17.07	21.10	25.98	29.50
SOFT-VQ [Che+25c]	0.59	29.48	7.48	1.64	5.63	10.52	15.19	18.97	21.93	24.13	25.38	25.93	26.07
MAE-TOK [Che+25b]	0.61	13.75	5.69	1.66	3.42	4.95	6.34	7.55	8.64	9.93	10.89	11.56	11.64
DE-TOK [Yan+26]	0.59	20.17	6.95	0.63	0.72	0.91	1.30	2.03	3.48	6.74	12.05	16.16	16.87
DM-VAE [Ye+25]	0.72	8.38	4.89	0.81	0.87	0.95	1.05	1.20	1.47	2.10	3.60	5.75	6.76
REPAE-VAE [Len+25]	0.56	25.75	6.46	0.94	1.78	3.10	4.91	7.08	9.86	13.12	17.18	21.04	22.14

Table 5: Quantitative results for 13 tokenizers. For each tokenizer, we evaluate rFID, gFID (w/o and w/ CFG), and GAR-FID without CFG at noise levels η_t ranging from 0.1 to 1.0. All results are obtained using SiT/XL models trained on the corresponding tokenizer [Xu+26].

Tokenizers	rFID ($\downarrow$)	gFID ($\downarrow$)		GAR-FID (η_t) ($\downarrow$)									
		w/o CFG	w CFG	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
SD-VAE [Rom+22]	0.73	25.91	6.33	1.11	1.76	2.91	5.03	8.27	12.82	17.31	20.43	21.85	22.30
IN-VAE [Len+25]	0.26	25.65	6.56	0.45	0.69	1.15	1.97	3.48	6.18	10.75	16.65	20.58	21.51
FLUX-VAE [Lab24]	0.16	34.06	7.82	0.96	1.66	3.13	5.54	8.87	13.45	18.62	24.00	27.85	29.65
QwenImage-VAE [Wu+25]	1.52	23.62	6.19	2.13	2.61	3.33	4.35	5.82	7.88	10.76	14.52	18.29	19.92
SD3-VAE [Ess+24]	0.21	26.38	6.31	0.59	0.90	1.51	2.59	4.50	7.38	11.62	16.91	21.27	22.89
EQ-VAE [Kou+25]	0.59	20.81	6.24	2.53	4.14	5.79	7.39	9.29	11.32	13.37	15.59	16.99	17.53
VA-VAE-16 [YW25]	0.30	8.57	4.20	2.11	2.75	3.30	3.85	4.32	4.88	5.49	6.30	7.32	7.80
VA-VAE-64 [YW25]	0.15	15.40	5.36	1.67	2.35	3.23	4.28	5.51	6.82	8.25	10.03	12.05	13.33
SOFT-VQ [Che+25c]	0.59	15.88	5.14	1.29	3.39	5.83	8.01	9.92	11.56	12.65	13.37	13.62	13.62
MAE-TOK [Che+25b]	0.61	6.27	3.74	1.23	1.99	2.59	3.08	3.54	3.94	4.44	4.91	5.19	5.21
DE-TOK [Yan+26]	0.59	11.97	4.83	0.65	0.73	0.88	1.13	1.52	2.26	3.77	6.58	9.14	9.59
DM-VAE [Ye+25]	0.72	4.65	3.39	0.81	0.86	0.92	1.00	1.09	1.24	1.50	2.12	3.17	3.86
REPAE-VAE [Len+25]	0.56	12.95	4.17	0.90	1.36	1.94	2.72	3.57	4.64	6.00	7.85	9.89	10.60
RAE [Zhe+26]	0.63	4.25	3.50	0.64	0.68	0.82	0.89	0.99	1.10	1.25	1.47	1.92	4.38

For each VAE, we use the publicly available SiT-B and SiT-XL [Ma+24] generative models released by iFID [Xu+26], which are trained on the 256×256 ImageNet dataset [Den+09] in the corresponding latent space. We directly use their provided checkpoints without additional retraining. For each tokenizer-generator configuration, we compute rFID, GAR-FID (without CFG) across noise levels $\eta_t \in [0.1, 1.0]$, and gFID both with and without classifier-free guidance (CFG). The results are summarized in Table 4 for SiT-B and Table 5 for SiT-XL.

Note on FID references. As shown in Table 4 and Table 5, GAR-FID (without CFG) at $\eta_t = 1.0$ is not equal to gFID without CFG. This discrepancy arises because the two metrics are evaluated with different reference image statistics. Specifically, gFID is computed using the precomputed ImageNet statistics (VIRTUAL_imagenet256_labeled.npz) [Den+09], whereas GAR-FID is computed against the 50K input-image PNG reference set used in the paired reconstruction protocol. As a result, GAR-FID at $\eta_t = 1.0$ is not expected to exactly match gFID without CFG.

To quantify the relationship between these metrics and generative performance, we follow iFID [Xu+26], and compute the Pearson correlation coefficient (PCC) and Spearman rank correlation coefficient (SRCC) between each metric and gFID under both settings (with and without CFG). As reported in Table 7, rFID exhibits consistently negative correlation with gFID, both with and without CFG, across SiT-B, SiT-XL, and their mixed configurations. This indicates that better reconstruction FID can correspond to worse generative FID, consistent with the reconstruction-generation dilemma observed in prior work [YW25; Kou+25; Ye+25; Sko+25; Che+25b; Xu+26]. In contrast, GAR-FID without CFG shows increasingly strong positive correlation with gFID as the noise level η_t increases. When $\eta_t \geq 0.7$, GAR-FID outperforms iFID [Xu+26] on both SiT-B and SiT-XL; when $\eta_t \geq 0.8$, it achieves near-perfect correlation with gFID across most settings.

The mixed SiT-B/XL setting further tests whether this correlation is preserved across generator scales. Correlations computed within a single generative model, such as SiT-B or SiT-XL, primarily measure how a metric responds to tokenizer variation under a fixed generator scale. By contrast, the mixed setting evaluates whether the metric

Table 6: List of VAEs included in our study. We follow iFID [Xu+26], and incorporate 13 VAEs with open-sourced models and train the corresponding diffusion models in their respective latent spaces.

VAE Name	Latent Dim.	Arch.	Training
SD-VAE [Rom+22]	$4 \times 32 \times 32$	UNet	Recon.
IN-VAE [Len+25]	$16 \times 16 \times 16$	UNet	Recon.
FLUX-VAE [Lab24]	$16 \times 32 \times 32$	UNet	Recon.
QwenImage-VAE [Wu+25]	$16 \times 32 \times 32$	UNet	Recon.
SD3-VAE [Ess+24]	$16 \times 32 \times 32$	UNet	Recon.
EQ-VAE [Kou+25]	$4 \times 32 \times 32$	UNet	Recon. + Equivariance
VA-VAE-16 [YW25]	$16 \times 16 \times 16$	UNet	Recon. + DINO alignment
VA-VAE-16 [YW25]	$64 \times 16 \times 16$	UNet	Recon. + DINO alignment
SOFT-VQ [Che+25c]	64×32	ViT	Recon. + DINO alignment
MAE-TOK [Che+25b]	128×32	ViT	Recon. + Mask + DINO alignment
DE-TOK [Yan+26]	128×32	ViT	Recon. + Mask + latent denoising
DM-VAE [Ye+25]	256×32	ViT	Recon. + Distribution Matching
REPAE-VAE [Len+25]	$4 \times 32 \times 32$	UNet	Recon. + REPA Loss
RAE [Zhe+26]	$768 \times 16 \times 16$	ViT	DINO Encoder + Recon. Decoder

Table 7: Correlation between FID-based metrics and generative performance (gFID). Correlations are computed across models obtained by training SiT with 13 pre-trained VAEs. We report both Pearson correlation (PCC) and Spearman rank correlation (SRCC) under different settings (with and without CFG) for SiT-B, SiT-XL, and their mixed configurations. GAR-FID without CFG consistently exhibits stronger correlation with gFID across noise levels η_t , whereas the standard reconstruction metric (rFID) shows negative correlation. **Bold** indicates the best results. [†] denotes results from [Xu+26], and * indicates our reproduced results.

Metrics	gFID SiT/B				gFID SiT/XL				gFID SiT/mixed			
	w/o CFG		w/ CFG		w/o CFG		w/ CFG		w/o CFG		w/ CFG	
	PCC (†)	SRCC (†)	PCC (†)	SRCC (†)	PCC (†)	SRCC (†)	PCC (†)	SRCC (†)	PCC (†)	SRCC (†)	PCC (†)	SRCC (†)
rFID [†]	-0.04	-0.31	-0.07	-0.31	-0.06	-0.21	-0.15	-0.31	—	—	—	—
iFID [†]	0.85	0.86	0.82	0.84	0.89	0.91	0.88	0.92	—	—	—	—
rFID*	-0.11	-0.36	-0.11	-0.30	-0.13	-0.34	-0.15	-0.39	-0.10	-0.30	-0.10	-0.28
iFID*	0.86	0.83	0.80	0.79	0.90	0.88	0.88	0.90	0.72	0.77	0.64	0.70
GAR-FID ($\eta_t = 0.1$)	-0.10	-0.19	-0.13	-0.14	0.01	0.00	0.10	0.01	0.04	0.03	0.05	0.01
GAR-FID ($\eta_t = 0.2$)	-0.04	-0.04	-0.08	-0.04	0.12	0.22	0.19	0.20	0.16	0.25	0.16	0.22
GAR-FID ($\eta_t = 0.3$)	0.14	0.15	0.09	0.14	0.25	0.38	0.32	0.35	0.34	0.42	0.34	0.39
GAR-FID ($\eta_t = 0.4$)	0.38	0.33	0.33	0.34	0.45	0.54	0.49	0.51	0.55	0.64	0.54	0.60
GAR-FID ($\eta_t = 0.5$)	0.63	0.68	0.57	0.69	0.61	0.66	0.65	0.61	0.74	0.80	0.72	0.76
GAR-FID ($\eta_t = 0.6$)	0.81	0.79	0.77	0.78	0.79	0.84	0.80	0.79	0.86	0.89	0.84	0.85
GAR-FID ($\eta_t = 0.7$)	0.93	0.90	0.89	0.87	0.91	0.90	0.91	0.86	0.95	0.96	0.93	0.91
GAR-FID ($\eta_t = 0.8$)	0.98	0.96	0.96	0.94	0.98	0.99	0.97	0.96	0.99	0.99	0.97	0.95
GAR-FID ($\eta_t = 0.9$)	1.00	0.99	0.98	0.98	1.00	1.00	0.98	0.96	1.00	1.00	0.98	0.97
GAR-FID ($\eta_t = 1.0$)	1.00	0.99	0.98	0.99	1.00	1.00	0.98	0.96	1.00	1.00	0.98	0.97

remains predictive when tokenizer choice and generator capacity vary jointly, which better reflects cross-system comparisons in latent visual generative modeling. In this more demanding regime, iFID shows weaker correlation with gFID, whereas GAR-FID exhibits slightly stronger correlation, indicating better robustness across tokenizer and generator variations. These results suggest that the predictive behavior identified in the controlled setting extends to more realistic cross-tokenizer and cross-generator comparisons, providing strong empirical evidence that GAR-FID is a robust proxy for generative performance.

H Decoder Feature Spaces for Latent Distribution Analysis in Section 3.1

In Section 3.1, we analyze intermediate representations from the SD-VAE decoder [Rom+22]. Specifically, we extract features from three representative spaces: (i) *Conv In*, the initial convolutional projection of latent inputs; (ii) *Mid Block*, the bottleneck residual block operating at the lowest spatial resolution; (iii) *Up Block 0*, the first upsampling stage in the decoder.

These feature spaces are identified based on the official implementation, and correspond to progressively deeper stages of the decoding process. We collect these activations using forward hooks at `decoder.conv_in`, `decoder.mid_block`, and `decoder.up_blocks[0]` for both encoder latents and generated latents. For visualization, decoder activations are adaptively pooled to a fixed 4×4 spatial resolution, flattened, and then projected using a one-dimensional LDA fitted to distinguish encoder-derived features from generation-derived features.

I gFID Evaluation Protocol and Implementation Details

Table 8: gFID is sensitive to the evaluation protocol. We compare two protocols on iMF-B/2: the official iMF evaluation [Gen+25b] and the commonly used OpenAI protocol. They differ in computational platform (TPU vs. GPU), Inception network implementation and weights (PyTorch vs. TensorFlow), and reference image statistics.

Setting	Platform	Inception	Reference Image Statistics	FID	IS
Official iMF	TPU	PyTorch	jit_in256_stats.npz	3.39	255.3
Official iMF (our reproduction)	GPU	PyTorch	jit_in256_stats.npz	3.37	255.7
OpenAI Setup	GPU	TensorFlow	VIRTUAL_imagenet256_labeled.npz	3.47	255.2

The official iMF-B/2 [Gen+25b] results are obtained under a different setup from the commonly used OpenAI evaluation protocol. Specifically, the two protocols differ in three aspects: (i) computational platform (TPU vs. GPU), (ii) Inception network implementation and weights (PyTorch vs. TensorFlow), and (iii) reference image statistics.

Table 8 summarizes these differences and their impact on reported FID scores. We observe that replacing the TPU platform with GPU, while keeping the other components unchanged, yields slightly improved FID in our reproduction. However, further adopting the OpenAI evaluation setup by switching to the TensorFlow Inception implementation and using different reference image statistics leads to noticeably worse FID. This indicates that the commonly used OpenAI evaluation protocol tends to produce systematically higher (worse) FID scores.

Unless otherwise specified, we adopt the commonly used OpenAI evaluation protocol (TensorFlow Inception network with VIRTUAL_imagenet256_labeled.npz statistics) for all reported gFID results, to ensure consistency and comparability across different models and prior works.

J Experimental Details in Section 7

Data Augmentation. All experiments are conducted on ImageNet-1k [Den+09]. Following LlamaGen [Sun+24], we apply iterative box downsampling to resize all images to a resolution of 256×256 .

Training Details. We perform decoder adaptation on four latent generative models, iMF-B/2, iMF-M/2, iMF-L/2, and iMF-XL/2 [Gen+25b], which share a common tokenizer, SD-VAE [Rom+22]. During adaptation, the tokenizer encoder and latent generative model are kept frozen, and only the tokenizer decoder parameters are updated. Generative models are performed without classifier-free guidance (CFG). All experiments are conducted on 8 NVIDIA H100 GPUs (80GB memory each) using the AdamW optimizer [LH19] with $\beta_1 = 0.9$ and $\beta_2 = 0.95$. We use a fixed learning rate of 10^{-5} and a weight decay of 10^{-4} , and train for 10 epochs. The batch size is set to 32 per GPU for all experiments. In terms of runtime, training typically completes within approximately 12 hours for iMF-B/2 and within 14 hours for iMF-XL/2 under the above settings.

Loss Weights and Noise Level. We fix $\lambda_1 = 1.0$, $\lambda_2 = 1.0$, $\lambda_3 = 0.7$, and $\lambda_4 = 10.0$ across all experiments. For noise level selection in decoder adaptation, we sample $\eta_t \sim \text{Uniform}[\eta_{\min}, \eta_{\max}]$, where $\eta_{\min} = 0.25$ and $\eta_{\max} = 0.45$, as discussed in Appendix E.

CFG parameters. The official iMF implementation employs three classifier-free guidance (CFG) parameters: $t_{\min}$, $t_{\max}$, and ω . In the original iMF repository, these parameters are selected via a fine-grained grid search. Since decoder adaptation is performed without CFG during training, the original CFG hyperparameters are no longer optimal after updating the decoder. We therefore conduct a new grid search to identify a new set of CFG parameters for evaluation. The updated CFG parameters are reported in Table 9. We will release the corresponding checkpoints to further facilitate reproducibility by the community.

K Additional Experimental Results in Section 7

We place additional experimental results in this appendix due to space constraints in the main paper. Specifically, we include reconstruction performance evaluations, extensive comparisons with a large number of additional baselines, and extensive qualitative comparisons.

Table 9: CFG parameters used for gFID evaluation across different model scales. Left: official parameters from the iMF repository, obtained via fine-grained grid search for models without decoder adaptation. Right: re-optimized parameters after decoder adaptation, obtained via a new grid search to account for changes in the decoder.

CFG Parameters	w/o Decoder Adaptation				w/ Decoder Adaptation			
	iMF-B/2	iMF-M/2	iMF-L/2	iMF-XL/2	iMF-B/2	iMF-M/2	iMF-L/2	iMF-XL/2
$t_{\min}$	0.40	0.40	0.40	0.42	0.50	0.55	0.44	0.48
$t_{\max}$	0.65	0.60	0.60	0.62	0.65	0.625	0.60	0.65
ω	8.0	10.5	10.5	8.0	9.0	10.5	8.0	5.25

Table 10: Reconstruction performance on the iMF-B/2 generative model. We compare reconstruction results without and with decoder adaptation. SR denotes standard reconstruction ($\eta_t = 0$), while GAR corresponds to generation-aware reconstruction with noise levels $\eta_t \in [0.1, 0.5]$.

Pipelines	w/o Decoder Adaptation					w/ Decoder Adaptation				
	PSNR ($\uparrow$)	SSIM ($\uparrow$)	LPIPS ($\downarrow$)	IS ($\uparrow$)	FID ($\downarrow$)	PSNR ($\uparrow$)	SSIM ($\uparrow$)	LPIPS ($\downarrow$)	IS ($\uparrow$)	FID ($\downarrow$)
SR ($\eta_t = 0.0$)	27.64	73.1	0.067	208.53	0.73	26.72	70.2	0.077	200.00	0.85
GAR ($\eta_t = 0.1$)	26.77	70.8	0.080	204.92	0.90	26.17	68.7	0.084	197.29	0.93
GAR ($\eta_t = 0.2$)	25.23	66.3	0.108	198.00	1.28	24.96	65.2	0.107	193.30	1.10
GAR ($\eta_t = 0.3$)	23.72	61.1	0.146	187.94	1.88	23.59	60.6	0.141	185.83	1.44
GAR ($\eta_t = 0.4$)	22.28	55.6	0.195	172.56	2.90	22.18	55.2	0.187	173.58	2.04
GAR ($\eta_t = 0.5$)	20.83	49.9	0.259	154.66	4.38	20.73	49.3	0.251	155.10	3.10

Reconstruction Results. As shown in Table 10, we compare reconstruction performance before and after decoder adaptation on the iMF-B/2 generative model, including peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), Fréchet Inception Distance [FID; Heu+17], perceptual similarity [LPIPS; Zha+18], and inception score [IS; Sal+16]. All results are reported without CFG. Standard reconstruction (SR, $\eta_t = 0$) degrades consistently after adaptation across all metrics, indicating that the adapted decoder is no longer well aligned with encoder latents z_e .

For generation-aware reconstruction (GAR), we observe a systematic trade-off in the effect of decoder adaptation as the noise level η_t increases. For structural fidelity, measured by PSNR and SSIM, decoder adaptation consistently degrades performance; however, the magnitude of degradation gradually diminishes with increasing η_t , indicating that the mismatch between the adapted decoder and the input latents becomes less severe as the latents move closer to the generative distribution. In contrast, perceptual quality, measured by LPIPS, IS, and FID, exhibits an opposite trend. At low noise levels, decoder adaptation slightly worsens performance, while at higher noise levels it yields consistent improvements. In particular, FID shows substantial gains when $\eta_t \geq 0.3$, suggesting that decoder adaptation improves perceptual alignment under generation-consistent latent distributions.

System-Level Comparison of Generative Performance. Table 11 presents a system-level comparison on class-conditional ImageNet 256×256 across diffusion/flow models with varying inference budgets (measured by the number of function evaluations, NFE), as well as representative methods from other generative model families. All results are reported with CFG when applicable. Under the most constrained setting of NFE = 1, applying decoder adaptation (DA) consistently improves generative performance across iMF model scales, with iMF-XL/2 + DA achieving the lowest gFID among all reported methods. This demonstrates that DA is highly effective under constrained inference budgets, particularly in the extreme low-compute setting. Compared to both from-scratch and distillation-based methods, our approach attains superior or competitive performance while maintaining minimal inference cost. These results highlight decoder adaptation as a lightweight yet effective mechanism for improving generative quality without increasing computational overhead.

Visual Comparison of Generated Samples. We provide additional qualitative comparisons for decoder adaptation (before and after) on iMF-XL/2 across 11 different classes, as shown in Figures 13–23. For each class, we present two sets of eight samples: the top row shows results before decoder adaptation, while the bottom row shows results after adaptation. Samples in the same column are generated from identical generative latents z_g , with the only difference being the decoder parameters. Across all classes, decoder adaptation consistently produces samples with richer fine-grained details, whereas samples before adaptation tend to appear slightly smoother. In addition, decoder adaptation reduces visual artifacts and yields more semantically consistent samples.

Table 11: System-level comparison on class-conditional ImageNet 256×256. Left: 1-NFE and 2-NFE diffusion/flow models trained *from scratch*. **Middle:** Diffusion/flow models, including distillation-based 1-NFE methods and multi-NFE methods. **Right:** Reference methods from other generative modeling families, including GANs and autoregressive/masking models. **NFE** denotes the number of function evaluations during inference. All numbers are with CFG when applicable, and ×2 in NFE indicates that the CFG computation doubles NFES at inference time. †: results cited from iMF [Gen+25b].

Method	# Params	NFE	FID	Method	# Params	NFE	FID	Method	# Params	NFE	FID
1-NFE diffusion/flow from scratch				1-NFE diffusion/flow (distillation)				GANs			
MeanFlow-XL/2 [†] [Gen+25a]	676M	1	3.43	π -Flow-XL/2 [†] [Che+25a]	675M	1	2.85	BigGAN [†] [BDS19]	112M	1	6.95
TiM-XL/2 [†] [Wan+25b]	664M	1	3.26	DMF-XL/2+ [†] [LYS26]	675M	1	2.16	GigaGAN [†] [Kan+23]	569M	1	3.45
α -Flow-XL/2+ [†] [Zha+25]	676M	1	2.58	FACM-XL/2 [†] [Pen+26]	675M	1	1.76	StyleGAN-XL [†] [SSG22]	166M	1	2.30
iMF-B/2 [†] [Gen+25b]	89M	1	3.39	Multi-NFE diffusion/flow				autoregressive/masking			
iMF-M/2 [†] [Gen+25b]	174M	1	2.27	ADM-G [†] [DN21]	554M	250×2	4.59	JetFormer-L [†] [TPK25]	2.75B	256×2	6.64
iMF-L/2 [†] [Gen+25b]	409M	1	1.86	LDM-4-G [†] [Rom+22]	400M	250×2	3.60	MaskGIT [†] [Cha+22]	227M	8	6.18
iMF-XL/2 [†] [Gen+25b]	610M	1	1.72	SimDiff [†] [HHS23]	2B	1000×2	2.77	RQ-	3.8B	256×2	3.80
iMF-B/2 + DA (Our)	89M	1	2.90	DiT-XL/2 [†] [PX22]	675M	250×2	2.27	Transformer [†] [Lee+22]			
iMF-M/2 + DA (Our)	174M	1	2.12	SiT-XL/2 [†] [Ma+24]	675M	250×2	2.06	STARFlow [†] [Gu+25]	1.4B	1024×2	2.40
iMF-L/2 + DA (Our)	409M	1	1.65	SiT-	675M	250×2	1.42	LLamaGen-3B [†] [Sun+24]	3.1B	256×2	2.18
iMF-XL/2 + DA (Our)	610M	1	1.56	XL/2 + REPA [†] [Yu+25]				VAR-d30 [†] [Tia+24]	2B	10×2	1.92
2-NFE diffusion/flow from scratch				SiD2 [†] [Hoo+25]	—	512×2	1.38	MAR-H [†] [Li+24]	943M	256×2	1.55
IMM-XL/2 [†] [ZES25]	675M	1×2	7.77	LightningDiT-XL/2 [†] [YW25]	675M	250×2	1.35	RAR-XXL [†] [Yu+24]	1.5B	256×2	1.48
MeanFlow-XL/2+ [†] [Gen+25a]	676M	2	2.20	DDT-XL/2 [†] [Wan+25a]	675M	250×2	1.26	xAR-H [†] [Ren+25]	1.1B	50×2	1.24
α -Flow-XL/2+ [†] [Zha+25]	676M	2	1.95	RAE [Zhe+26] + DiT ^{OH} -XL [†]	839M	250×2	1.13				
iMF-XL/2 [†] [Gen+25b]	610M	2	1.54								



Figure 13: Class 14: indigo bunting (top: before, bottom: after).



Figure 14: Class 42: agama (top: before, bottom: after).



Figure 15: Class 81: ptarmigan (top: before, bottom: after).

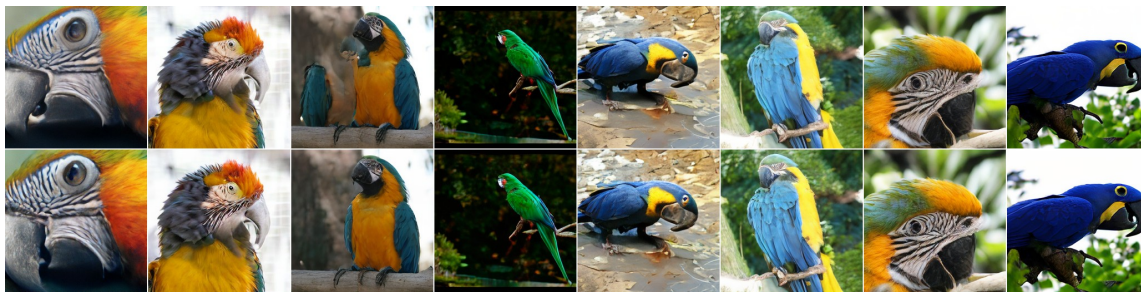


Figure 16: Class 88: macaw (top: before, bottom: after).

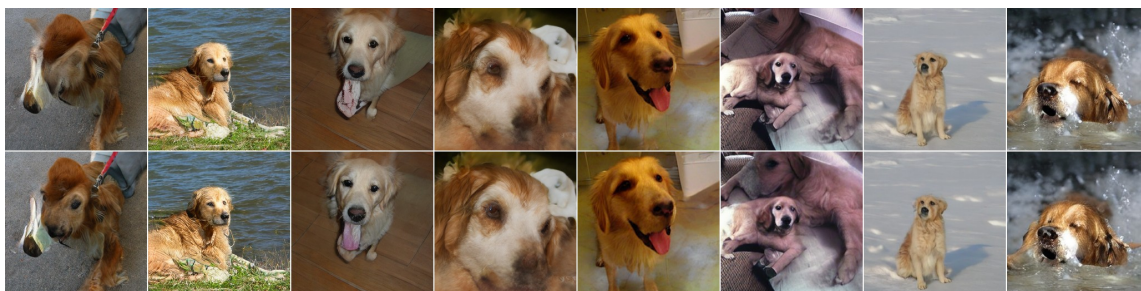


Figure 17: Class 207: golden retriever (top: before, bottom: after).

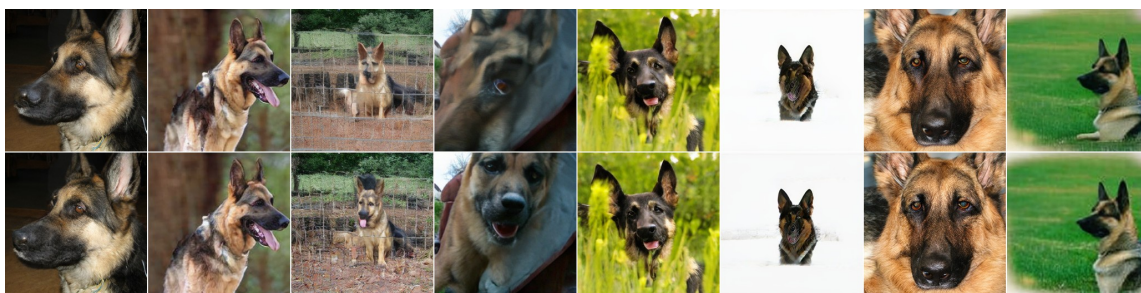


Figure 18: Class 235: German shepherd (top: before, bottom: after).

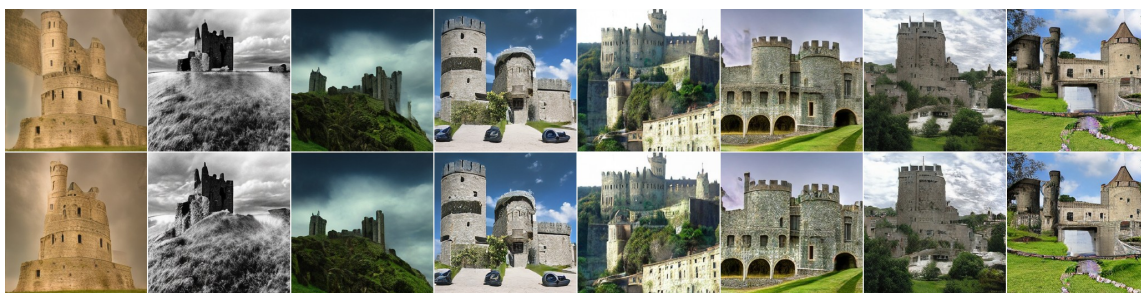


Figure 19: Class 483: castle (top: before, bottom: after).

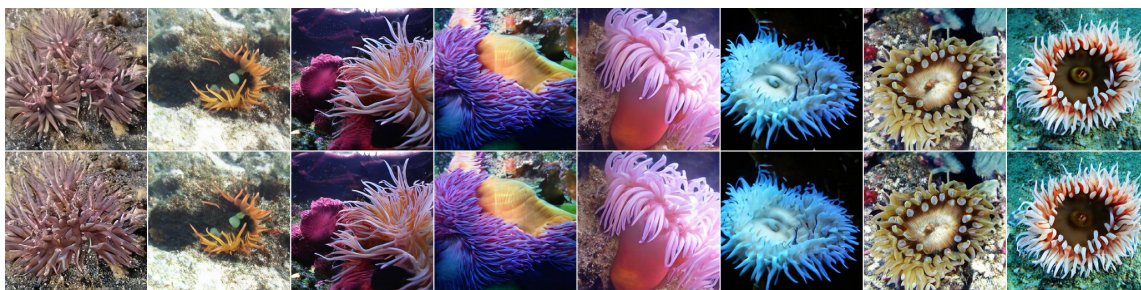


Figure 20: Class 108: sea anemone (top: before, bottom: after).

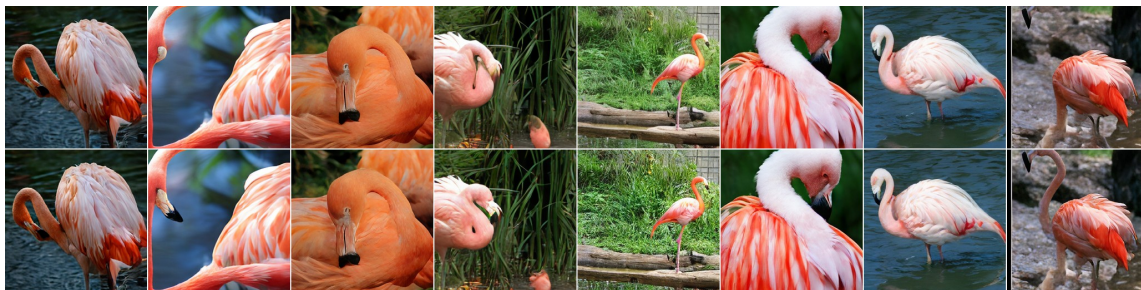


Figure 21: Class 130: flamingo (top: before, bottom: after).



Figure 22: Class 22: bald eagle (top: before, bottom: after).



Figure 23: Class 817: sports car (top: before, bottom: after).